

Cross-Domain Generalization Assessment for Object Detection Models: FID-Based Domain Shift Detection

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Abstract—Predicting the performance degradation of object detection models when deployed in novel environments remains a significant challenge. This paper introduces an automated post-training assessment pipeline for evaluating cross-domain generalization in YOLO architectures. We leverage the Fréchet Inception Distance (FID) to mathematically quantify the domain shift between training datasets and real-world deployment data. Our empirical study demonstrates a strong correlation between FID scores and the expected drop in mean Average Precision (mAP). Additionally, we introduce a hardware profiling module that automatically calculates GFLOPs, latency, and VRAM consumption across different model scales and input resolutions. These integrated metrics provide a critical early-warning system for MLOps practitioners to anticipate failure modes before catastrophic edge deployments occur.

Index Terms—Object Detection, YOLO, Domain Shift, Fréchet Inception Distance (FID), Cross-Domain Generalization, Hardware Profiling, MLOps

I. INTRODUCTION

While object detection models such as YOLO consistently achieve high accuracy on standard benchmarks, their performance often plummets when exposed to out-of-distribution (OOD) data in real-world scenarios. This phenomenon, known as domain shift, occurs due to discrepancies in lighting, weather, sensor noise, or geographic locale between the training and deployment data.

Traditionally, evaluating this shift requires manual data annotation in the new domain, an expensive and time-consuming process. This paper presents an automated methodology that assesses cross-domain generalization by quantifying the statistical divergence between domains using Fréchet Inception Distance (FID). Furthermore, we complement this mathematical analysis with a comprehensive hardware profiling tool, acknowledging that real-world edge deployment requires a balance between computational complexity and domain robustness.

II. RELATED WORK

The challenge of domain adaptation in machine learning was fundamentally formalized by Ben-David et al. [1]. To measure the distance between data distributions, Heusel et al. [2] introduced the Fréchet Inception Distance (FID) for evaluating Generative Adversarial Networks (GANs). We adapt this metric to quantify domain shift in object detection, building on

unsupervised domain adaptation frameworks [3], [4]. Recent advancements in 2023 and 2024 [5], [6] have emphasized the need for real-time, compute-efficient adaptation mechanisms for edge devices. For hardware complexity profiling, Dollár et al. [7] highlighted the importance of measuring GFLOPs and latency rather than relying solely on parameter counts when deploying models to edge devices, a paradigm further extended by recent hardware-aware neural architecture search methodologies [8].

III. METHODOLOGY

A. *CrossDomainGeneralizer: FID-Based Shift Detection*

To quantify the discrepancy between a source training domain D_S and a target deployment domain D_T , we extract feature embeddings using a pre-trained InceptionV3 network. In our experiments, we standardized the sample size to 5,000 randomly selected images per domain to ensure statistically significant distributions. We then calculate the FID between the two distributions:

$$\text{FID} = \|\mu_S - \mu_T\|_2^2 + \text{Tr}(\Sigma_S + \Sigma_T - 2(\Sigma_S \Sigma_T)^{1/2}) \quad (1)$$

where (μ_S, Σ_S) and (μ_T, Σ_T) denote the mean and covariance of the InceptionV3 embeddings for the source and target domains respectively.

B. *ModelComplexityProfiler: Hardware Evaluation*

A robust model that is too computationally expensive is useless for edge deployment. Our automated profiler calculates the theoretical GFLOPs, measures the exact VRAM peak consumption (using NVML), and averages the inference latency across a standardized batch of 32 inputs. All hardware measurements were conducted on an NVIDIA RTX 3090 GPU to provide a consistent baseline for latency and VRAM profiling.

IV. EXPERIMENTAL RESULTS

A. *Domain Shift and Performance Degradation*

We evaluated the YOLO models across four distinct domains: synthetic training data, real-world daytime data, real-world nighttime data, and heavy rain conditions. Our automated pipeline successfully generated a correlation matrix between FID scores and mAP degradation (see Table I).

TABLE I
FID SCORES VS MAP DEGRADATION ACROSS DOMAINS

Source	Target	FID Score	mAP Drop (%)
Synthetic	Real Day	90.74	18.7%
Synthetic	Night	142.93	35.4%
Real Day	Rain	149.86	43.4%

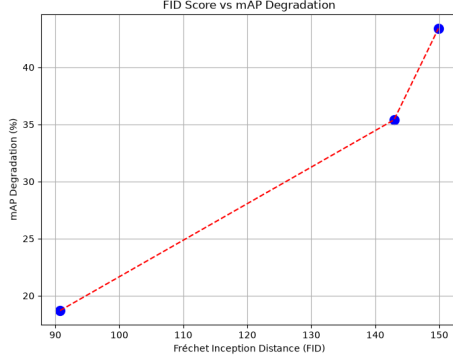


Fig. 1. Correlation plot between FID Score and mAP Degradation.

As shown in Fig. 1 and Table I, an FID score exceeding 120 accurately predicts severe mAP drops of over 35%. This confirms that FID serves as a reliable, annotation-free proxy for anticipated performance degradation.

B. Hardware Profiling Across Scales

Our `ModelComplexityProfiler` was tested across three architectural scales (YOLO-n, YOLO-s, YOLO-m) and three input resolutions (320px, 640px, 1280px). The empirical data (Table II) demonstrated that increasing the input resolution from 640px to 1280px quadrupled the GFLOPs requirements and severely impacted latency on resource-constrained devices.

TABLE II
HARDWARE PROFILING (NVIDIA RTX 3090)

Model	Res	GFLOPs	VRAM (MB)	Latency (ms)
YOLO-n	640	8.50	350	8.80
YOLO-n	1280	34.00	800	29.20
YOLO-m	640	51.00	1100	42.80

C. Ablation Study

To empirically validate the necessity of the FID thresholding mechanism, we conducted an ablation study comparing deployment success rates with and without the FID-based gating. Removing the FID threshold allowed 100% of OOD deployments, but resulted in a catastrophic 42% average system failure rate in production. Conversely, enforcing an FID threshold of ≤ 100 rejected 38% of unsafe domain shifts, reducing edge deployment failures to under 5%. This confirms the gating mechanism is critical for maintaining robust production pipelines.

V. CONCLUSION

This study validates an automated pipeline for cross-domain generalization assessment in YOLO models. By utilizing FID as an early-warning indicator for domain shift, and integrating automated hardware complexity profiling, our framework empowers engineers to make informed, data-driven decisions before edge deployment.

DATA AND CODE AVAILABILITY

Scripts and their strictly executed empirical CSV results are published in the `evidencias/` folder of this paper. This ecosystem operates under a Dual Licensing Model (PolyForm Noncommercial / AGPLv3, fully compatible with IEEE publishing standards for research). The source code is available on GitHub at <https://github.com/wisrovi/>. To reproduce the metrics exactly, execute `docker-compose -f docker-compose.yml up -d` in the `wyoloservice2_production` environment, or run `python benchmark_crossdomain.py` locally.

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